EARTH OBSERVATION

FloodSight AI

Comprehensive Operations & User Manual

Evidence-Based Satellite Mapping, Verification Gates, Live Radar & Inundation Intelligence

SYSTEM VERSION:	1.0.0-sen1floods11	**POLICY INTEGRITY:**	ZERO-FABRICATION GUARANTEED
CHALLENGE TRACK:	Track B: Mapping Flood Damage From Space	**RELEASE DATE:**	October 2026

Executive Documentation Summary

FloodSight AI transforms raw multi-spectral satellite imagery into verifiable flood extent, transportation network exposure, and community isolation intelligence. This document serves as the authoritative operational guide for emergency response personnel, geospatial specialists, and incident command logisticians operating the platform.

Verifiable Ingestion

Pre-inference verification of Sentinel-2 MSI GeoTIFF headers, bands, and EPSG CRS tags.

Fail-Closed Guardrails

Zero synthetic outputs; returns HTTP 503 if model weights are missing.

Live Radar & Forecasts

Real-time RainViewer radar mosaics with playback and Open-Meteo precipitation.

ICS-209 Reporting

Standardized operational incident briefings grounded exclusively in measured telemetry.

OPERATIONAL SCOPE & ETHICAL NOTICE

FloodSight AI is a decision-support platform designed to assist emergency management agencies. Satellite flood masks detect surface water accumulation and must be combined with ground reconnaissance for structural collapse verification.

Table of Contents

Chapter 1: Executive Overview & System Mission

Chapter 2: Interface Architecture & Getting Started

Chapter 3: Interactive 3D Earth & Global Geographic Navigation

Chapter 4: Satellite Raster Verification & Flood Segmentation

Chapter 5: Tactical Geospatial Maps & Live Environmental Layers

Chapter 6: 3D Topographic Simulation & Digital Elevation Modeling

Chapter 7: Incident Command Briefings & Situation Reports (SITREPs)

Chapter 8: Metric Formulations, Geospatial Calculations & Verification Rules

Chapter 9: Troubleshooting, Error Reference & Safe Recovery

Chapter 10: Data Governance, Security Architecture & Scientific Integrity

Chapter 11:

Quick Reference Appendices (Checklist, Glossary & API Matrix)

CHAPTER 1

1.1 Purpose & Target Users

FloodSight AI is a specialized Earth-observation intelligence platform developed for Track B: Mapping Flood Damage From Space. It processes high-resolution multi-spectral satellite imagery (Sentinel-2 MSI) to determine pixel-level flood extent, infrastructure disruption, and community isolation during catastrophic hydrological events.

- **Incident Commanders & EOC Directors:** Formulate operational triage decisions and resource allocations based on verified cartographic evidence.
- **Geospatial & Remote Sensing Engineers:** Inspect coordinate reference systems (CRS), ground sampling distances (GSD), and band alignments.
- **Field Logisticians:** Identify accessible transportation corridors, submerged bridges, and stranded population centers.

1.2 System Requirements

The web application is accessible on modern desktop command center browsers with WebGL 2.0 hardware acceleration enabled:

- **Recommended Browsers:** Google Chrome 110+, Microsoft Edge 110+, Mozilla Firefox 115+, Apple Safari 16.4+
- **Hardware:** Dedicated GPU or modern multi-core SoC (Apple Silicon, Intel Iris Xe, AMD Radeon, NVIDIA GeForce) for real-time 3D Earth rendering.
- **Network Bandwidth:** Minimum 5 Mbps broadband for satellite raster uploads and real-time RainViewer radar tile streams.

1.3 Zero-Fabrication Commitment

MANDATORY SCIENTIFIC POLICY: ZERO FABRICATION

FloodSight AI strictly prohibits synthetic, fabricated, or simulated measurements. If a model checkpoint is absent, inference is refused with an explicit HTTP 503 error. All metrics are derived from authenticated data sources.

CHAPTER 2

2.1 Persistent Navigation Bar

The top navigation bar provides instant access to all core platform workspaces, status indicators, and emergency tools:

- **Overview (H):** 3D global Earth visualization, observation target camera flights, and platform architecture overview.
- **Command Hub (D):** Central operations center reporting system health, model readiness, and recent raster analyses.
- **Analysis (A):** Satellite imagery ingestion portal, cryptographic verification gate, and U-Net segmentation engine.
- **Tactical Maps (M):** High-resolution Leaflet and Google maps featuring live RainViewer radar, Open-Meteo forecasts, and flood extent overlays.
- **Reports (R):** ICS-209 compliant incident situation reports for 20 global disasters or your active uploaded analysis.
- **Methodology (I):** Comprehensive 8-stage scientific workflow documentation and dataset contracts.

2.2 Command Center Global Keyboard Shortcuts

For high-tempo emergency operations, single-key shortcuts allow immediate workspace transitions without mouse usage:

Key	Destination Workspace	Functionality Description
H	Overview / Landing	Planetary Earth globe, target selector, and 3D simulation
D	Command Hub	System health status, recent chip telemetry, quick workflows
A	Analysis Laboratory	Upload GeoTIFF rasters, run verification, execute inference
M	Tactical Maps	Live weather radar, Google tactical maps, 3D terrain
R	Reports & SITREPs	Operational incident briefings and situation reports
I	Methodology	Dataset contracts, scientific pipeline documentation
?	Shortcuts Modal	Displays the global interactive keyboard shortcuts help
Esc	Dismiss / Close	Closes modals, floating cards, and active menus

CHAPTER 3

3.1 3D Globe Navigation (Orbit, Pan, Zoom)

The planetary Earth visualization on the Overview workspace is rendered via Three.js with multi-spectral surface textures, atmospheric scattering shaders, and physical planetary rotation:

- **Orbit Navigation:** Left-click + drag orbits the planet smoothly around its polar axis.
- **Zoom Altitude:** Scroll wheel or trackpad pinch zooms between orbital view (z = 4.2) and continental altitude (z = 2.0).
- **Camera Controls:** Zoom In (+), Zoom Out (-), and Reset Globe View buttons located in the bottom-right corner.

3.2 Observation Targets Console

The Observation Targets selector features 20 pre-validated disaster incidents from the global registry (e.g., Pakistan Sindh, USA Kentucky, Bangladesh Sylhet). Clicking any target triggers a smooth 700ms cubic-eased camera flight directly to the coordinates.

> **OPERATIONAL NOTICE: ILLUSTRATIVE GLOBE VS VERIFIED MAP**
> The 3D globe provides geographic discovery and situational orientation. The glowing hotspot rings represent incident epicenters. Tactically verified flood extents and evacuation routes must be inspected on the Tactical Map.

CHAPTER 4

4.1 Model Input Contract (s2-optical-v1)

FloodSight AI enforces an explicit model input specification to ensure reproducible, scientifically defensible segmentation:

Parameter	Contract Requirement	Technical Rationale
Sensor Platform	ESA Sentinel-2 MSI	Calibrated against Sen1Floods11 splits
File Format	GeoTIFF (.tif, .tiff)	Requires valid TIFF magic headers
Bands & Order	3 Bands: B4 (Red), B3 (Green), B2 (Blue)	Optical multi-spectral water contrast
Tile Dimensions	Exactly 512 x 512 pixels	Neural network tensor input requirement
Ground Resolution	10.0 meters / pixel	Native Sentinel-2 optical ground sampling distance
Coordinate Reference	Embedded EPSG tags, ModelPixelScaleTag	Mandatory for physical geodesic hectarage

4.2 Pre-Inference Server Verification Gate (/api/verify-image)

All rasters must pass server-side verification before entering the inference pipeline. The gateway checks:

- **File Integrity:** Validates non-empty buffer and decodable TIFF/BigTIFF magic headers.
- **Spatial Dimensions:** Verifies width and height equal exactly 512x512 pixels.
- **Modality & Channels:** Validates 3-band optical mapping and EPSG projection tags.
- **Verification Token:** Registers a cryptographically random UUIDv4 token in the server SafeTokenRegistry with a 1-hour TTL.

4.3 Fail-Closed Inference Gate (/api/analyze)

MODEL CHECKPOINT GUARDRAIL

If no trained model checkpoint exists at models/flood_segmentation/floodsight_unet_resnet34_best.pth, inference returns HTTP 503 FLOOD_MODEL_UNAVAILABLE. No synthetic masks or placeholders will ever be generated.

CHAPTER 5

5.1 Live Weather Radar & Precipitation Loops

The Tactical Maps workspace incorporates real-time radar mosaics from RainViewer and weather forecasts from Open-Meteo:

- **Radar Timeline:** Provides the past 2 hours of observed radar reflectivity at 10-minute intervals, plus 30 minutes of nowcast predictive frames.
- **Playback Controls:** Play/Pause animation, Skip Back, Skip Forward, and adjustable playback speed (500ms to 2000ms per frame).
- **Color Palettes:** Select between Universal Blue (Default), RainViewer Classic, and Titan Spectrum.

5.2 Meteorological Telemetry & Hourly Forecasts

Live queries to Open-Meteo display real-time micro-climate indicators for the selected coordinate center:

- **Current Telemetry:** Temperature (°C), Relative Humidity (%), Wind Speed (km/h), Wind Direction (degrees), Surface Pressure (hPa), Precipitation Rate (mm/h).
- **48-Hour Rainfall Forecast:** Visual precipitation chart predicting rainfall accumulation over the next two days to project crest timing.

OPERATIONAL WARNING: WEATHER RADAR VS. SATELLITE INUNDATION

Weather radar indicates atmospheric rain in the clouds. Satellite optical/SAR imagery indicates standing floodwater on the ground. Never dispatch rescue boats based solely on radar rain reflectivity.

CHAPTER 6

6.1 Interactive 3D Terrain Viewer

Rendered via Three.js / React Three Fiber, the 3D terrain viewer models representative regional elevation geometries:

- **Extrusion Scale Slider (1.0x - 5.0x):** Exaggerates vertical elevation relief to emphasize subtle drainage basins, ridges, and natural levees.
- **Water Level Slider (0.0 - 1.0):** Simulates rising water column stage heights, demonstrating how valleys and lowlands become inundated.
- **Procedural Disclosure:** Representative terrain models are procedural approximations designed for hydraulic stage intuition, not LiDAR-grade engineering surveys.

CHAPTER 7

7.1 ICS-209 Incident Situation Reports

The Reports workspace formats verified satellite metrics into standardized Incident Command System briefings:

- **Global Disaster SITREPs:** Pre-configured situation reports for 20 benchmarked global flood disasters.
- **Active Upload Intelligence:** Auto-generated briefing based exclusively on your freshly uploaded satellite scene.
- **Zero-Fabrication Guardrail:** If road vectors or demographic census datasets are absent, reports explicitly state "DATASET MISSING" rather than estimating numbers.

CHAPTER 8

8.1 Mathematical Formulations

Every metric displayed on FloodSight AI is computed through deterministic mathematical formulas:

- **Flood Coverage Fraction:** f = N_flood / N_valid, where N_valid represents non-NoData pixels.
- **Mean Confidence Score:** Average positive-class probability across all pixels classified as water (P >= threshold).
- **Projected Hectarage (UTM):** Area (ha) = (N_flood * Delta_x * Delta_y) / 10,000, where Delta_x and Delta_y are in meters.
- **Geodesic Hectarage (WGS84):** Calculated via ellipsoidal arc distance formulas scaled by local latitude cosine factors.

CHAPTER 9

9.1 Diagnostic Error Matrix

Error Code / Symptom	Root Cause Diagnosis	Safe Operator Recovery Action
REJECTED: Magic Header	Corrupted file or non-TIFF format	Re-export as uncompressed GeoTIFF (.tif)
MODEL_DIMENSION_MISMATCH	Raster size is not 512x512	Resample or tile the image to 512x512 pixels
HTTP 503 FLOOD_MODEL_UNAVAILABLE	Model weights missing on server	Install trained checkpoint at models/flood_segmentation/
HTTP 403 VERIFICATION_REQUIRED	Token expired (1 hour TTL)	Click "Run Integrity Verification" to re-authenticate
STALE DATA / SOURCE UNAVAILABLE	RainViewer / Open-Meteo timeout	Click "Refresh Telemetry"; previous data remains viewable
WebGL Context Lost	Hardware acceleration disabled	Enable "Use hardware acceleration" in browser settings

CHAPTER 10

10.1 In-Memory Token Lifecycle (SafeTokenRegistry)

Verification tokens are maintained in-memory on the backend gateway with strict safeguards:

- **Automatic TTL Pruning:** Verification records automatically expire after 1 hour (3,600,000 ms).
- **Memory Capacity Cap:** Maximum 1,000 records maintained simultaneously to prevent memory exhaustion.
- **Privacy Assurance:** Uploaded satellite scenes are analyzed in local memory and are not uploaded to third-party cloud buckets.

CHAPTER 11

11.1 Operational Analysis Checklist

Follow this sequential procedure for every satellite analysis mission:

- **Step 1: Check System Health:** Ensure Command Hub (D) reports HEALTHY with trained model checkpoint detected.
- **Step 2: Format Satellite Scene:** Ensure GeoTIFF is 512x512 pixels, 3 bands (B4/B3/B2), with embedded EPSG georeferencing.

- **Step 3: Run Server Verification:** Upload file in Analysis (A) and verify green VERIFIED_SATELLITE_INPUT status.
- **Step 4: Select Operating Threshold:** Set probability threshold slider (default: 0.50).
- **Step 5: Execute U-Net Segmentation:** Click Execute U-Net Flood Analysis and inspect coverage % and physical hectarage.
- **Step 6: Review Situation Briefing:** Switch to Reports (R) -> Active Upload to inspect the standardized incident briefing.

11.2 API Gateway Reference

Endpoint Path	HTTP Method	Function & Return Codes
/api/health	GET	System status, model availability (200 OK)
/api/verify-image	POST	Inspects GeoTIFF headers, issues verification token (200 OK, 400, 415)
/api/analyze	POST	Executes U-Net segmentation (200 OK, 403 Forbidden, 503 Unavailable)
/api/external/weather-radar	GET	Proxies live RainViewer radar frames (200 OK, 502, 504 Timeout)
/api/external/weather-forecast	GET	Proxies Open-Meteo precipitation forecast (200 OK, 400, 504)
/api/external/hydrology	GET	GloFAS river discharge telemetry (200 OK / disclosed nulls)